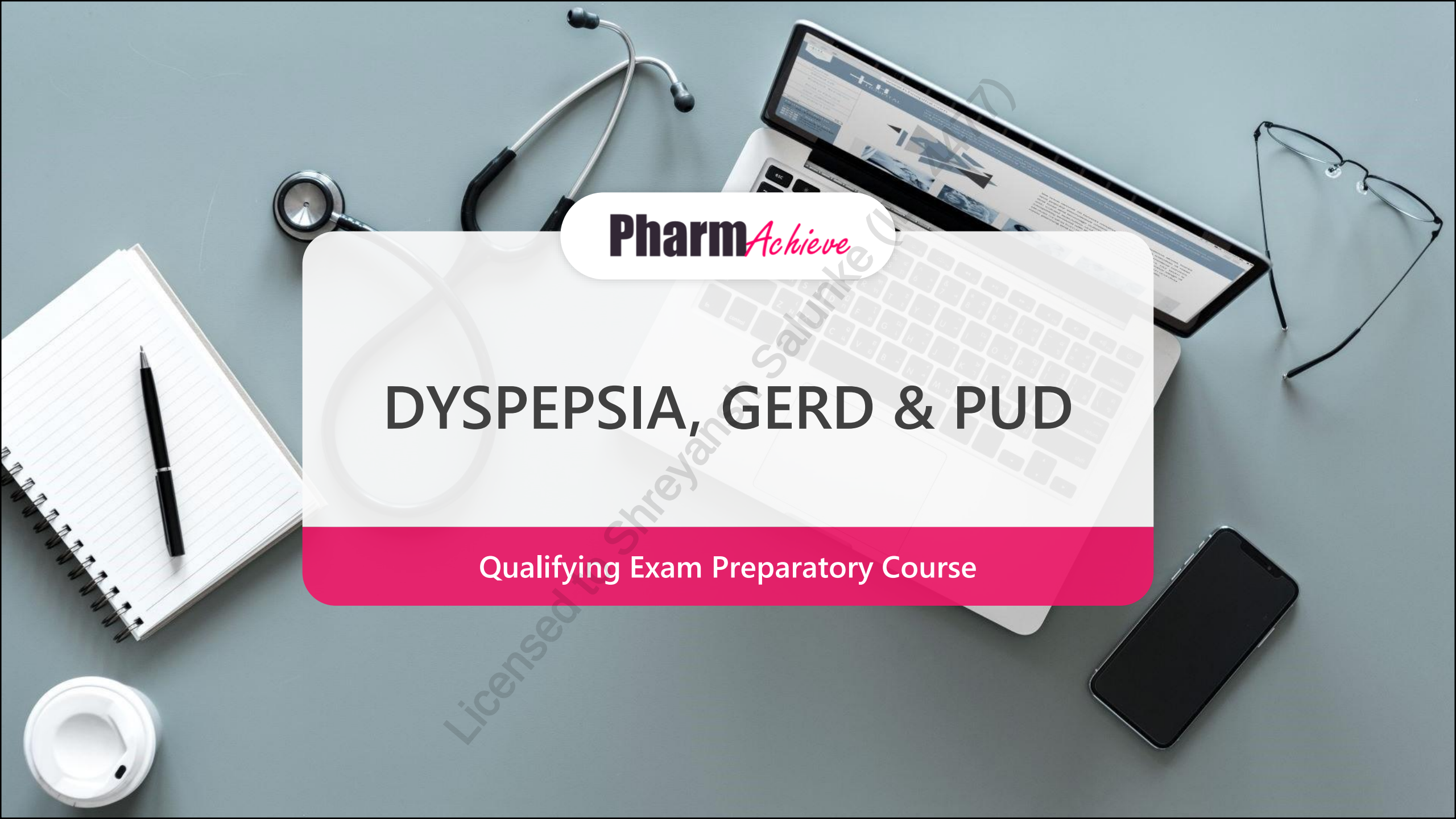




Pharm*Achieve*

DYSPEPSIA, GERD & PUD

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

Abx	Antibiotics
AC	Before Meals
AEs	Adverse Events
Al	Aluminum
ASA	Acetylsalicylic Acid (Aspirin)
CCB	Calcium Channel Blocker
BID	Twice Daily
Ca	Calcium
C/I	Contraindication
COPD	Chronic Obstructive Pulmonary Disease
Cox-2	Cyclo-oxygenase 2
D/C	Discontinue
D/I	Drug Interactions
DR	Delayed-release
ED	Erectile Dysfunction
GCS	Glucocorticoid
GERD	Gastroesophageal Reflux Disease
GIT	Gastrointestinal Tract
<i>H. pylori</i>	<i>Helicobacter pylori</i>
H2RA	Histamine 2 Receptor Antagonist
IV	Intravenous

Mg	Magnesium
MOA	Mechanism of Action
NSAID	Non-steroidal Anti-Inflammatory Drug
OTC	Over the Counter
PAC	PPI + Amoxicillin + Clarithromycin
PAL	PPI + Amoxicillin + Levofloxacin
PAM	PPI + Amoxicillin + Metronidazole
PAMC	PPI + Amoxicillin + Metronidazole + Clarithromycin
PAR	PPI + Amoxicillin + Rifabutin
PBMT	PPI + Bismuth + Metronidazole + Tetracycline
PC	After Meals
PMC	PPI + Metronidazole + Clarithromycin
PO	By mouth
PPI	Proton Pump Inhibitor
PRN	As Needed
PUD	Peptic Ulcer Disease
QID	Four Times Daily
SOB	Shortness of Breath
Sx	Symptoms
Tx	Treatment

DEFINITIONS

Gastroesophageal reflux disease

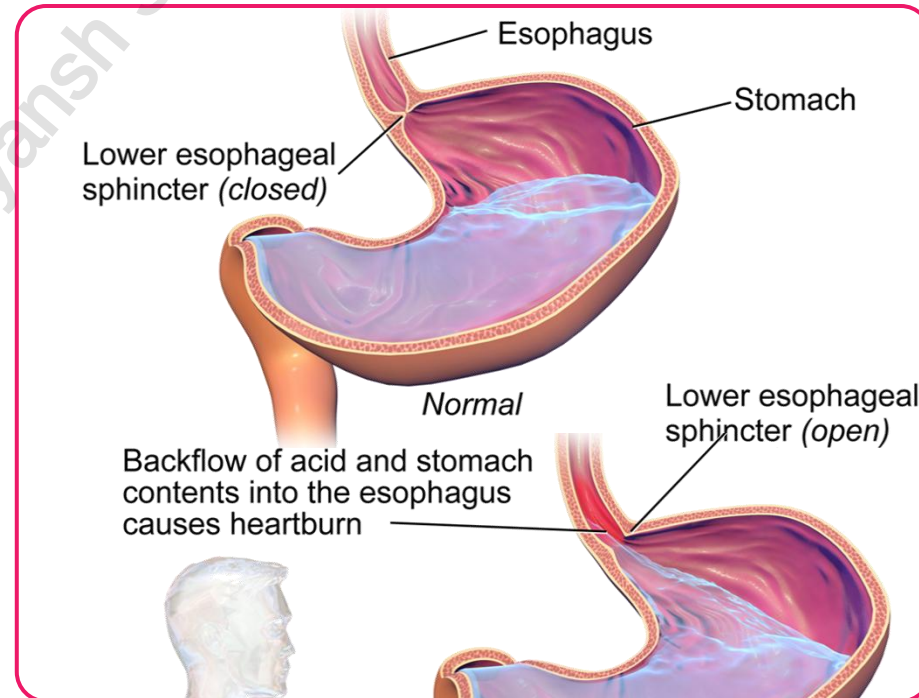
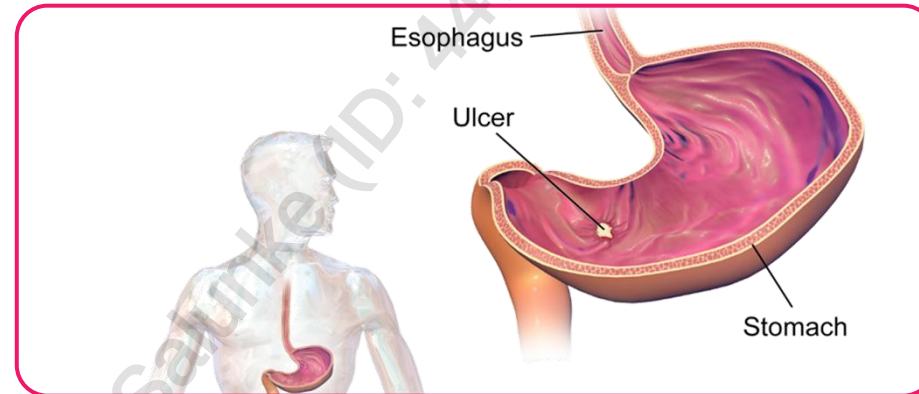
- Heartburn and regurgitation due to an excessive reflux of gastric contents

Peptic ulcer disease

- Development of ulcers/erosions of the mucosa, often leading to dyspeptic symptoms and/or bleeding

Dyspepsia

- Chronic, recurrent epigastric pain or discomfort in the upper abdomen for > 1-month, early satiety



CLINICAL PRESENTATION

Peptic Ulcer Disease	Dyspepsia	GERD
• Dyspepsia • Nausea, vomiting • Belching, bloating, distension • Tarry stools	• Epigastric pain • Belching, bloating, distension • Early satiety • Nausea, vomiting, heartburn (infrequent)	• Many patients are asymptomatic • Predominant heartburn +/- regurgitation • Hypersalivation • Belching • Exacerbated by bending over, eating high-fat meals

PATHOPHYSIOLOGY OF PEPTIC ULCER DISEASE

An insult or injury to the gastric or intestinal mucosa: various causes: H. pylori infection, drug-induced (NSAIDs, chronic steroids, antiplatelet/anticoagulants, smoking, alcohol)



The injury does not heal adequately



Erosion of the gastric/intestinal mucosal lining



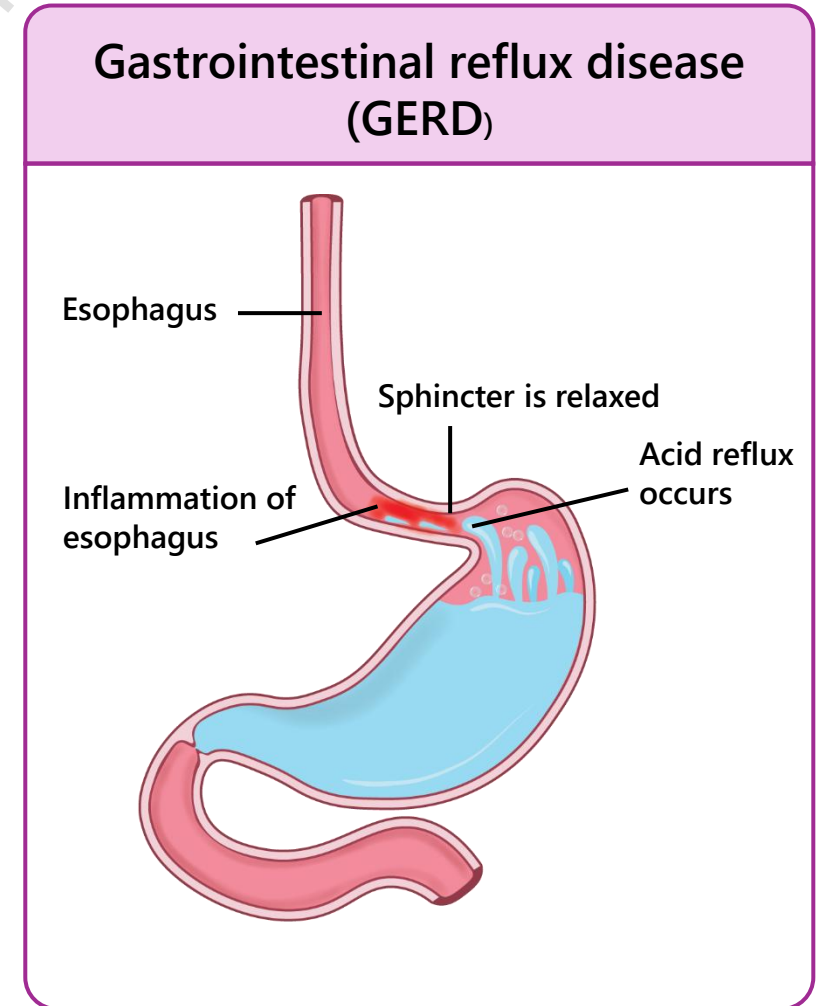
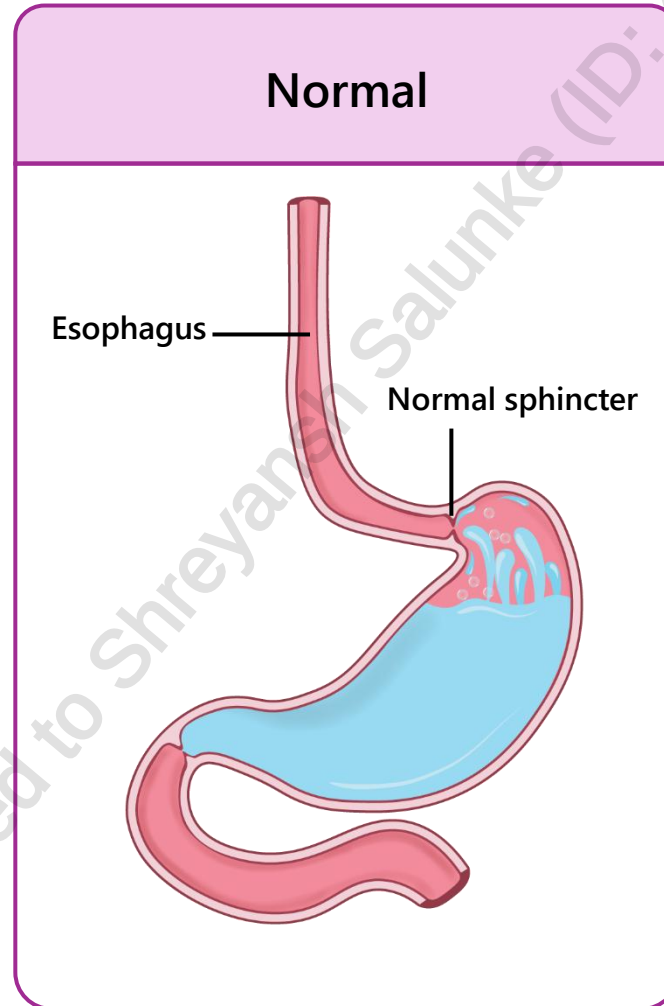
Ulcer formation



Bleeding

PATHOPHYSIOLOGY OF GASTROESOPHAGEAL REFLUX DISEASE

- Factors that cause a **weakened lower esophageal sphincter tone** will promote the backflow of stomach acid into the esophagus resulting in burning sensation, regurgitation, globus sensation, difficulty swallowing



RISK FACTORS

Overall risk factors	Risk factors for PUD	Risk factors for GERD
• Older age >65 years • Lifestyle: smoking, obesity, fatty foods, alcohol, coffee, chocolate, cannabis • Medical conditions including COPD, stress, anxiety	• History of PUD or GI bleeding • <i>H. pylori</i> infection • Medication: NSAIDs, corticosteroids, antiplatelets, bisphosphonates • Medical Conditions: Zollinger-Ellison syndrome, chronic gastritis, delayed gastric emptying • Certain foods	• Medication: alpha blockers, CCBs, opioids, anticholinergics, benzodiazepines, nicotine • Medical Conditions: obesity, pregnancy, hiatus hernia, delayed gastric emptying, impaired esophageal peristalsis, hypersecretory conditions, bile reflux

BARRETT'S ESOPHAGUS

- Change in cell type in epithelium of GIT due to chronic GERD
- Can predispose patients to esophageal cancer

Endoscopy should be used to screen for Barrett's esophagus in patients who have:

- Family history of Barrett's esophagus or esophageal adenocarcinoma
- Chronic symptoms of GERD PLUS one of the following risk factors:
 - ✓ Male sex
 - ✓ >50 years
 - ✓ Caucasian
 - ✓ Central obesity
 - ✓ History of smoking

GOALS OF THERAPY

- ✓ Decrease or eradicate symptoms
- ✓ Decrease or prevent recurrence
- ✓ Heal damaged mucosa
- ✓ Prevent complications (e.g. Barrett's esophagus, esophageal adenocarcinoma)
- ✓ Minimize side effects associated with drug therapy

NON-PHARMACOLOGICAL MEASURES

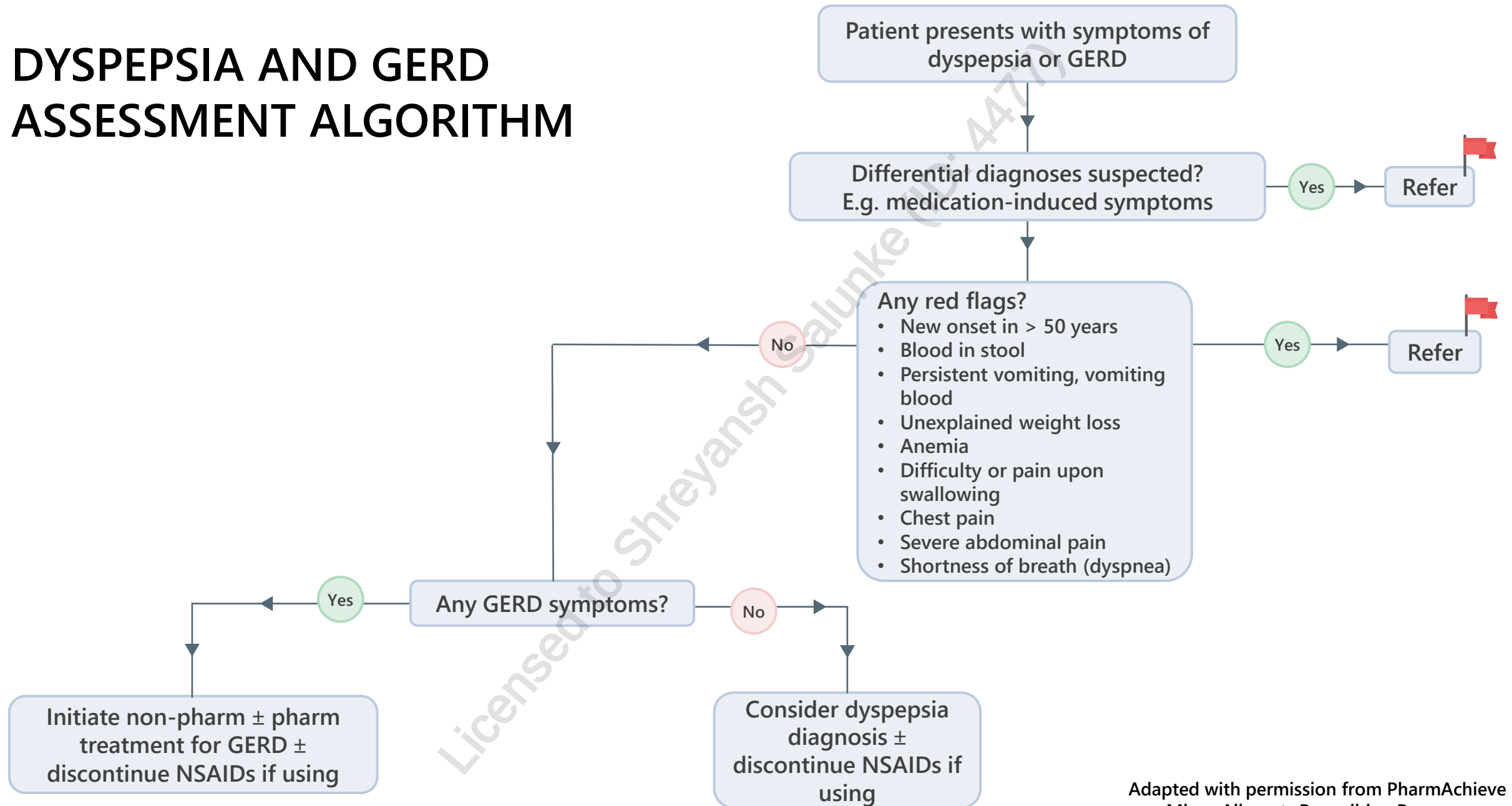
Dyspepsia and peptic ulcer disease

- Avoid foods that aggravate dyspepsia: caffeine, acidic/spicy/fatty foods and drinks
- Avoid chronic, high-dose ASA or NSAIDs
- Drugs that may cause or aggravate dyspepsia: bisphosphonates, codeine, calcium channel blockers, metronidazole, iron salts, beta-adrenergic agonists, tricyclic antidepressants and anticholinergic medications

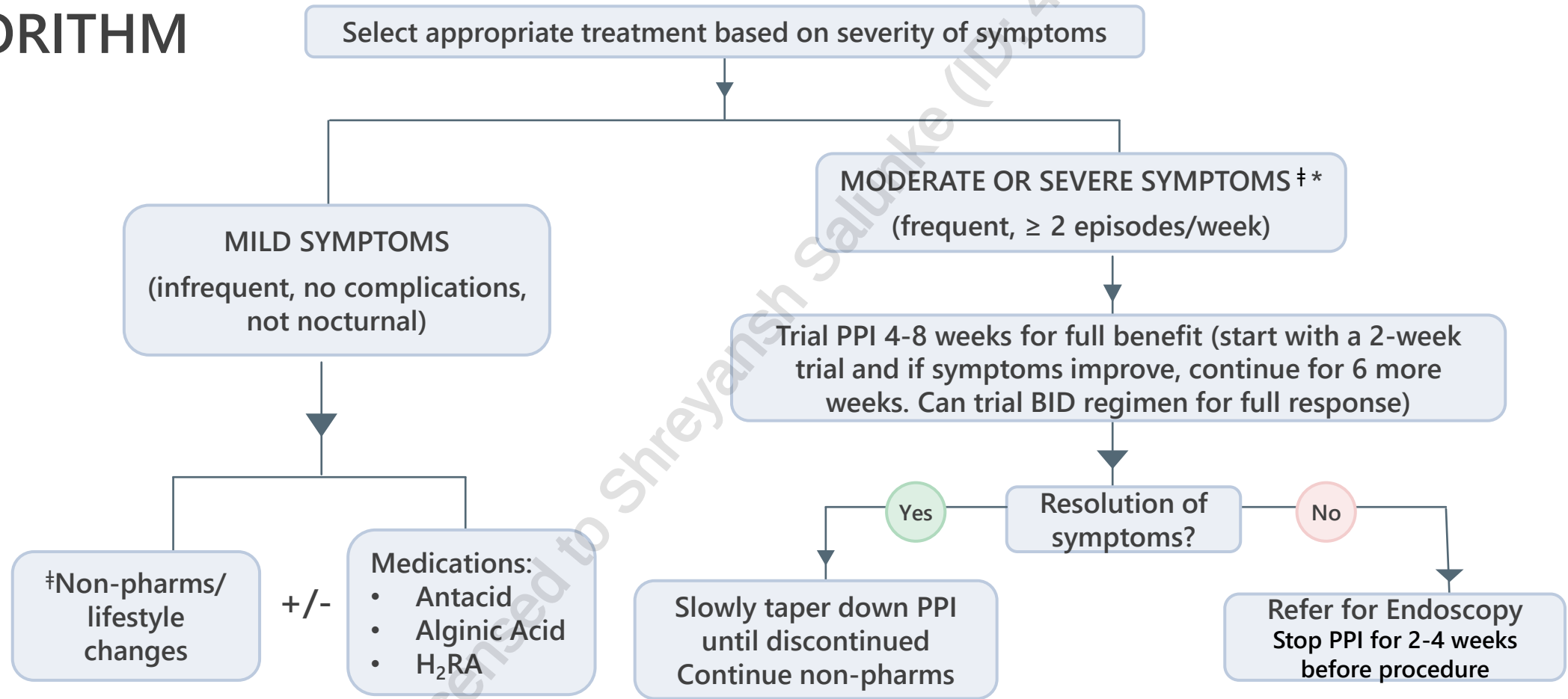
Gastroesophageal reflux disease

- Avoid trigger foods
- Eat smaller, more frequent meals
- Avoid tight fitting clothes around waistband
- Avoid meals 3 hours before bedtime, especially if nocturnal GERD
- Refrain from lying down immediately after meals
- Weight loss
- Smoking/alcohol cessation
- Elevate the head of the bed by 4-6" if nocturnal GERD

DYSPEPSIA AND GERD ASSESSMENT ALGORITHM



GERD TREATMENT ALGORITHM

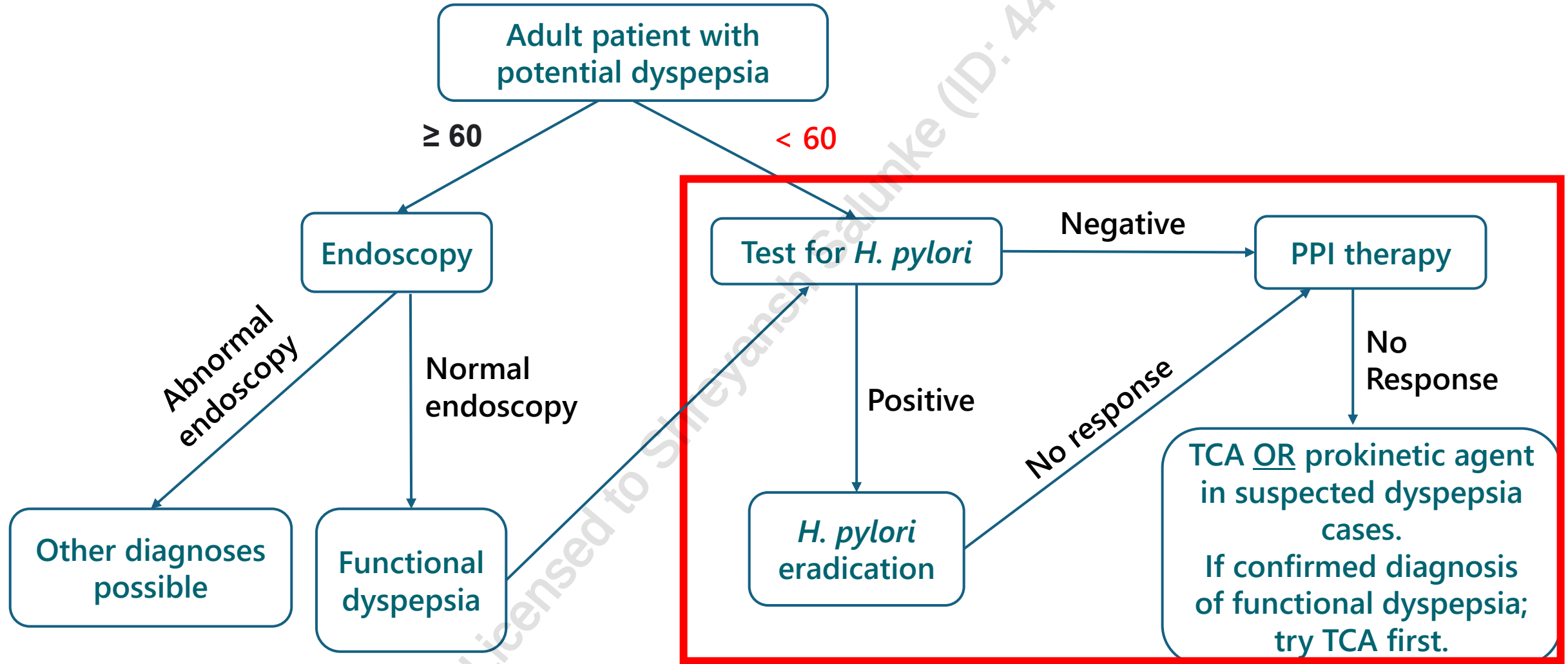


‡ Non-pharms/lifestyle changes should be encouraged for moderate-severe GERD as well

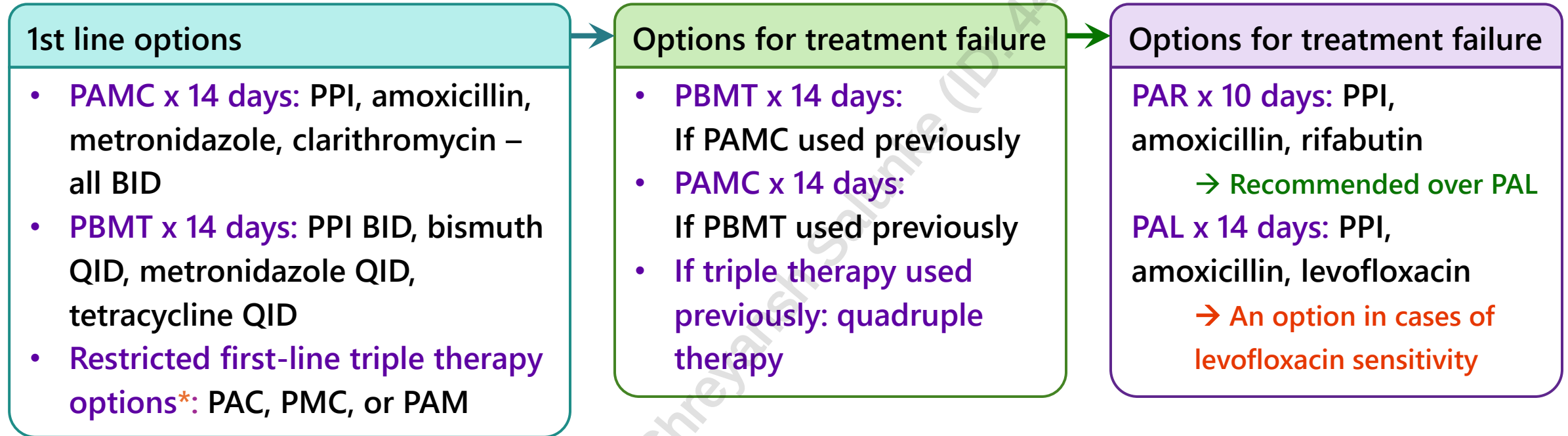
*May add adjunctive antacid, alginic acid or H₂RA for symptomatic relief

Adapted with permission from PharmAchieve
Minor Ailments Prescribing Program

DYSPEPSIA TREATMENT ALGORITHM



H. PYLORI TREATMENT ALGORITHM



*Restricted to areas with known low clarithromycin resistance (<15%) or proven local eradication rates (>85%)

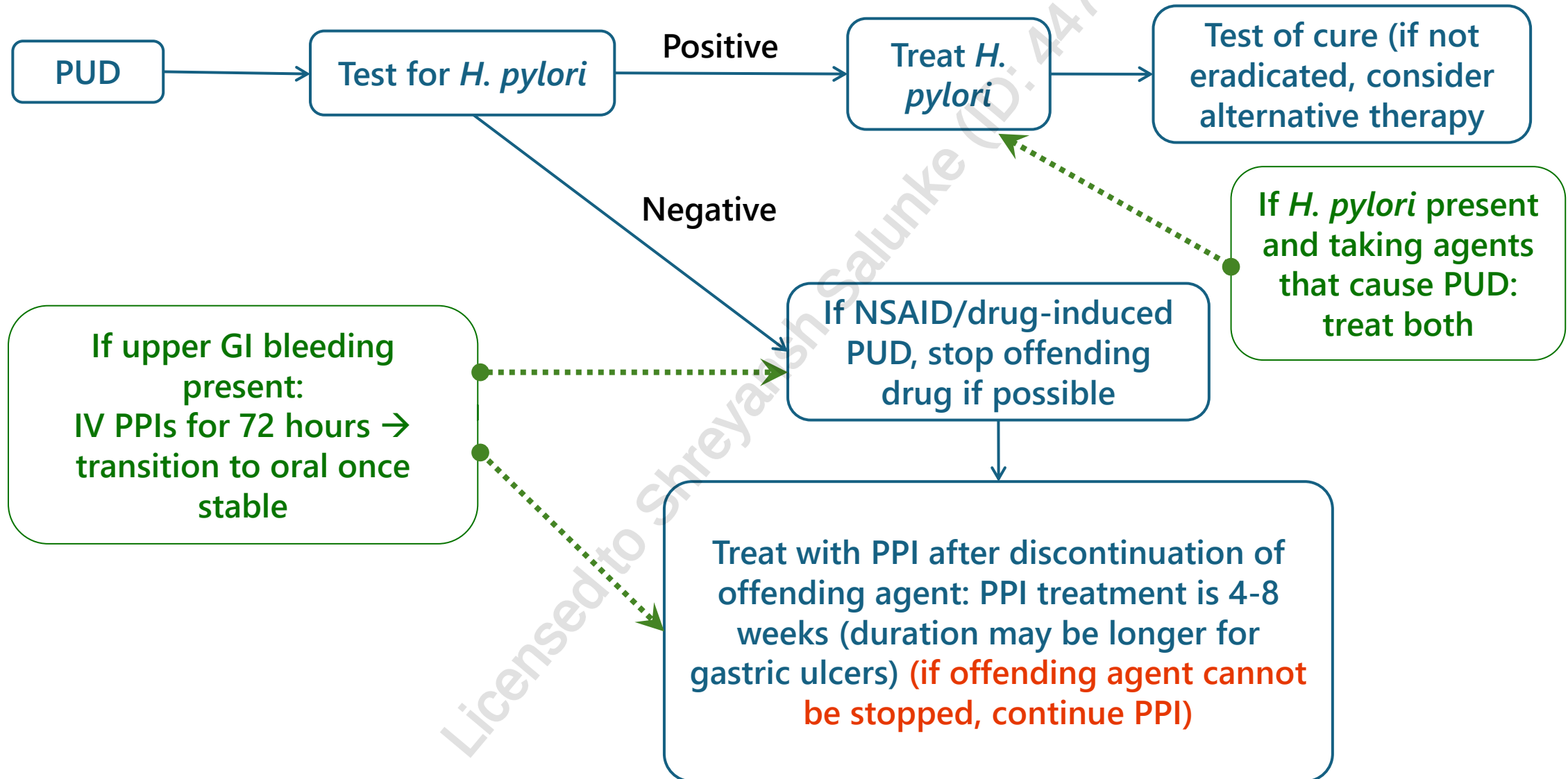
Treatment Pearls:

- ❑ PAMC should not be used in patients who have failed clarithromycin therapy or have clarithromycin-resistant strains
- ❑ PBMT should be used as first-line therapy in patients with penicillin allergy
- ❑ PAR should be used as third-line therapy over PAL except in the case of levofloxacin sensitivity

***H. PYLORI* TREATMENT DOSES**

Drug	Dose
Amoxicillin	1 g BID
Bismuth subsalicylate	524 mg QID
Clarithromycin	500 mg BID
Levofloxacin	500 mg BID
Metronidazole in PAMC	500 mg BID
Metronidazole in PBMT	500 mg QID
Tetracycline	500 mg QID
Rifabutin	150 mg BID

PEPTIC ULCER DISEASE TREATMENT ALGORITHM



PROTON PUMP INHIBITORS

All PPIs have similar efficacy

Agents	Safety	Interactions
Dexlansoprazole, lansoprazole, omeprazole, pantoprazole, rabeprazole, esomeprazole	- Side effects: nausea, diarrhea, headache, rash - Chronic effects: B12 and Mg deficiency, enteric infection risk (<i>C. Diff</i>, gastroenteritis), osteoporosis, fundic gland polyps	- Omeprazole, esomeprazole, pantoprazole, and lansoprazole are CYP2C19 inhibitors - Medications that require acidic environment may not be absorbed

COMMENTS

- **1st line** for GERD, dyspepsia, PUD, & GI Bleeds
- Used concomitantly with chronic NSAID use → PUD prevention
- PPI > H₂RA efficacy in healing esophagitis + recurrence prevention
- Take 30-60 min before meals; Dexlansoprazole can be taken with or without food
- May try to taper patients off their PPI if on long-term (6 months or more)
- If the patient is taking clopidogrel → pantoprazole, rabeprazole, lansoprazole and dexlansoprazole are preferred. Omeprazole/esomeprazole should be avoided with clopidogrel (level X interaction)

HISTAMINE-2 RECEPTOR ANTAGONISTS

Agents	Safety	Interactions
Cimetidine, famotidine, ranitidine	Side effects: headache, dizziness, drowsiness, constipation, diarrhea, nausea, vomiting, tachycardia, rash; gynecomastia (<4%) and ED <i>with high-dose, long-term cimetidine</i>	• Cefuroxime PO • Risedronate DR • Itraconazole • Ketoconazole • Cimetidine has many D/I compared to others • Famotidine least anticholinergic
COMMENTS • Can be used in GERD & PUD; longer duration than antacids • Mild GERD: can be used PRN; <i>better efficacy for nocturnal GERD</i> than meal-related GERD • Tachyphylaxis occurs rapidly		

ANTACIDS AND ALGINIC ACID

Agents	Safety	Interactions
- Antacids: calcium carbonate, aluminum/magnesium hydroxide - Alginates	- Antacids side effects: diarrhea (magnesium-containing products), constipation (aluminum/calcium-containing products), discoloration - Alginic acid side effects: nausea, vomiting, flatulence, eructation	Interferes with absorption of medications (e.g. fluoroquinolones, bisphosphonates, levothyroxine, amphetamines)

COMMENTS

- Indicated for short-term symptomatic relief of heartburn/indigestion GERD*
- Normally administered after meals
- Avoid magnesium and aluminum-containing products in renal dysfunction
- Tips:** Antacids should be separated from bisphosphonates by at least 30 minutes, fluoroquinolones by 2-6 hours (2 hours if quinolones are taken before antacids and 6 hours if taken after), and levothyroxine by at least 4 hours
 - Bisphosphonates are usually taken first thing in the morning, 30-60 minutes before food or other medications

MISOPROSTOL

Agent	Safety	Interactions
Misoprostol	Side effects: headache, abdominal pain, diarrhea, constipation, flatulence, nausea/vomiting, uterine cramping, vaginal bleeding	- Increased diarrhea risk when used with magnesium-containing antacids - C/I: pregnancy (acts as an abortifacient)
COMMENTS - Prevention of NSAID-induced <i>gastric</i> ulcers causing PUD - Less effective than PPI standard dose for ulcer treatment Take with food and avoid magnesium-containing antacids		

SPECIAL POPULATIONS- PREGNANCY AND BREASTFEEDING

Pregnancy

- Antacids, H2RA, alginates and PPIs are considered safe in pregnancy
 - Exception - sodium bicarbonate can cause fluid retention and metabolic alkalosis
- Do NOT use misoprostol
- If H. pylori positive, delay treatment until after pregnancy and breastfeeding (treat as uninvestigated dyspepsia)



Breastfeeding

- Antacids, alginates, ranitidine and PPIs may be used



MONITORING PARAMETERS

EFFICACY ENDPOINTS

Parameter	Degree of Change	Timeframe
Symptoms (e.g. pain, burning, etc.)	Improved and/or resolved	Days-weeks
GI bleeding	Prevent/resolve	Ongoing
Ulceration	Healed	4-12 weeks
Complications and/or Recurrence	Prevent	Ongoing

SAFETY ENDPOINTS

Parameter	Degree of Change	Timeframe
Medication AEs	Prevent or minimize	Ongoing

TAKEAWAY

GERD - Predominant heartburn +/- regurgitation, worsened by bending or high-fat meals

- Mild symptoms → non-pharms/ lifestyle changes +/- antacids, H₂RA or alginic acid
- Moderate - severe symptoms → trial PPI 4-8 weeks. If no symptom improvement, refer for endoscopy. Recommend non-pharms/lifestyle changes.

Dyspepsia - Uncomfortable indigestion with fullness, bloating, burping, nausea, and epigastric pain

- ≥ 60 years → Endoscopy
- < 60 years → Test for *H. pylori* → Eradicate if positive. If negative, start PPI
- ***H. pylori* treatment:** PAMC or PBMT as first and second line therapies. PAR is recommended over PAL as a third-line option, except in the case of levofloxacin sensitivity

PUD – Dyspepsia, acid reflux, tarry stools

- Test for *H. pylori* → Eradicate if positive. If negative, stop offending agent and treat with PPI
- With **clopidogrel**, use preferred PPIs (pantoprazole, rabeprazole, lansoprazole, dexlansoprazole) instead of omeprazole or esomeprazole

CASE 1

GD is a 34-year-old female who comes into your pharmacy seeking advice. She explains she has been experiencing heartburn throughout the day over the last few weeks. She has never had this before. She states that when she lies down for bed, it is worse. GD is 32 weeks pregnant, and she has been craving fried foods, which seem to make her symptoms worse.

Apart from her pregnancy, her medical history is significant for depression and seasonal allergies. GD currently takes prenatal vitamins. She stopped her antidepressant (fluoxetine 20 mg po daily) 8 months ago out of concern for the effects on her baby. She does not smoke, use recreational drugs or consume alcohol. This is her first pregnancy, and she is concerned about taking anything that may harm the baby.



CASE 1 - QUESTIONS

1. Based on GD's symptoms, she most likely is suffering from which of the following?
 - a) PUD
 - b) GERD
 - c) *H. pylori*
 - d) Dyspepsia
2. Which of the following would be the most appropriate recommendation?
 - a) Lifestyle modifications + pantoprazole
 - b) Ranitidine
 - c) Lifestyle modifications + antacids
 - d) Watch & wait
3. Which of the following is NOT an appropriate non-pharmacological option for GD?
 - a) Avoid fried foods
 - b) Avoid eating 3 hours before bedtime
 - c) Weight loss
 - d) Eat smaller, more frequent meals
4. GD asks you why she is experiencing this when she has never had heartburn before. Which of the following is the LEAST appropriate explanation?
 - a) Increased intra-abdominal pressure
 - b) Progesterone relaxes the lower esophageal sphincter thus permitting reflux
 - c) The later the stage of pregnancy, the more likely GERD will occur
 - d) Discontinuation of her anti-depressant (fluoxetine)

CASE 2

PD is a 56-year-old male who has been complaining of persistent pain below his ribs over the last 2 months. Upon being asked, PD says he smokes approximately 15 cigarettes per day as he has been incredibly stressed at work lately. PD confirms he has not had any recent weight loss or blood in his stool. PD has a medical history significant for anxiety, dyslipidemia and osteoporosis. He takes escitalopram 20 mg po daily, rosuvastatin 40mg po daily and alendronate 70mg po once weekly. He also takes vitamin D 2000 units po daily for general health. He is allergic to penicillin which causes anaphylaxis.

- 1) Which of the following would be the LEAST appropriate first step?
- a) Inquire how he takes his alendronate
 - b) Refer PD for an endoscopy to rule out malignancy
 - c) Refer PD for a urea breath test to rule out *H. pylori*
 - d) Advise PD that smoking can increase his risk of peptic ulcer disease and esophageal cancer, and inquire whether he would be interested in a discussion on smoking cessation



CASE 2 - QUESTIONS

2) PD was tested for *H. pylori* and the results came back positive. Which of the following would be an appropriate first-line treatment for him?

- a) PPI + Bismuth + Metronidazole + Tetracycline
- b) PPI + Amoxicillin + Metronidazole + Tetracycline
- c) PPI + Bismuth + Doxycycline
- d) PPI + Clindamycin + Cephalexin

3) Which of the following is true?

- a) *H. pylori* treatment duration is 10 days, and PPI is dosed once daily
- b) *H. pylori* treatment duration is 14 days, and PPI is dosed twice daily
- c) *H. pylori* treatment duration is 10 days, and PPI is dosed twice daily
- d) *H. pylori* treatment duration is 14 days, and PPI is dosed once daily

4) PD therapy for *H. pylori* includes a PPI. Which of the following is false?

- a) PPIs can increase infection risk with long-term use
- b) PPIs are not as effective as H₂RA in the treatment of *H. pylori* and GERD
- c) PPIs have the same overall efficacy when compared to one another
- d) Pantoprazole is best taken 30-60 minutes prior to eating a meal

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CHANGE LOG

August 2025

- Slide 3 & Slide 17: Removed abbreviation 'OD' as per ISMP's Do Not Use List
- Slide 3: Removed abbreviations 'RF', 'PG' & 'HF' as they are not in the lecture
- Slide 4: Updated definition of dyspepsia to better reflect 2018 AJG guidelines (added > 1 month)
- Slide 9: Updated screening criteria to better reflect guidelines (male sex is one of the risk factors considered, but women can also be screened)
- Slide 10: Added Barrett's esophagus + esophageal adenocarcinoma as potential complications
- Slide 13: Removed pregnancy from red flags
- Slide 16: Updated PAR ranking to third line (now preferred over PAL), added treatment pearls
- Slide 25: Edited Takeaway
- Slide 26-27: Reworded and changed questions to be more complex
- Slide 28-29: Added details and changed questions to make it more complex
- Slide 30: Added references

November 2025

- Formatting and phrasing changes made throughout
- Slide 11: Updated legend
- Slide 13: Removed 'patients < 18 years' from red flags
- Slide 14: Added that non-pharms should be recommended for all patients with GERD
- Slide 16: Added frequency of medications
- Slide 24: Updated degree of change of GI bleeding
- Slide 25: Updated Takeaway
- Slide 30: Updated references

March 2026

- Slide 9: Updated information
- Slide 12: Added persistent vomiting to red flags
- Slide 13: Updated algorithm of moderate-severe symptoms
- Slide 16: Updated levofloxacin dose to BID
- Slide 17: Minor updates to algorithm
- Slide 20: Updated comments on antacids
- Slide 27: Changed option 4 for clarity